time analysis

Cailin Wilson

2026-05-04

Are certain types of crime more likely to occur at specific times of day or days of the week?

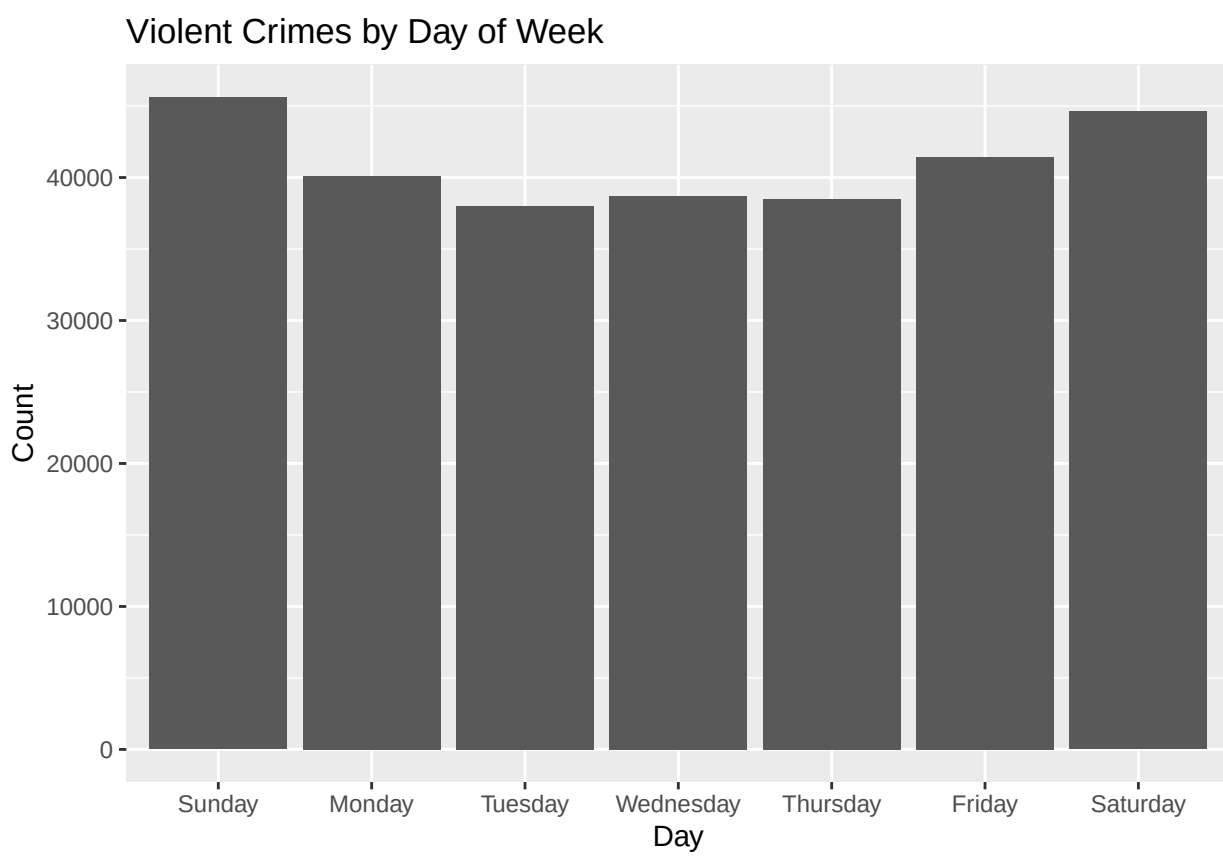
```
df <- read_csv("Fixed_Date_version.csv")
```

```
unique_codes <- unique(df[, c("Crm Cd", "Crm Cd Desc")])

unique_codes <- unique_codes[order(unique_codes$`Crm Cd`), ]

violent_codes <- c(
  110, 113, 121, 122, 210, 220, 230, 231, 235, 236,
  250, 251, 434, 435, 436, 437, 622, 623, 624, 625,
  626, 627, 647, 753, 755, 756, 760, 761, 763, 810,
  812, 815, 820, 821, 822, 860, 910, 920, 921, 922,
  926, 928, 930, 940, 943
)

violent_df <- df %>%
  mutate(
    crm_cd = as.numeric(`Crm Cd`)
  ) %>%
  filter(crm_cd %in% violent_codes)
```

```
violent_df <- violent_df %>%
  mutate(

    # day of week
    date_only = as.Date(`DATE OCC`),

    weekday = wday(
      date_only,
      label = TRUE,
      abbr = FALSE
    ),

    # hour of day from TIME OCC
    time_chr = str_pad(
      as.character(`TIME OCC`),
      width = 4,
      pad = "0"
    ),
```

```
    hour = as.numeric(
      substr(time_chr, 1, 2)
    )
  )
```

```
day_counts <- violent_df %>%
  count(weekday)

ggplot(day_counts,
       aes(x = weekday, y = n)) +
  geom_col() +
  labs(
    title = "Violent Crimes by Day of Week",
    x = "Day",
    y = "Count"
  )
```




```
day_max <- day_counts %>%
  slice_max(n, n = 1)

day_min <- day_counts %>%
  slice_min(n, n = 1)

cat(
  "Highest:",
```

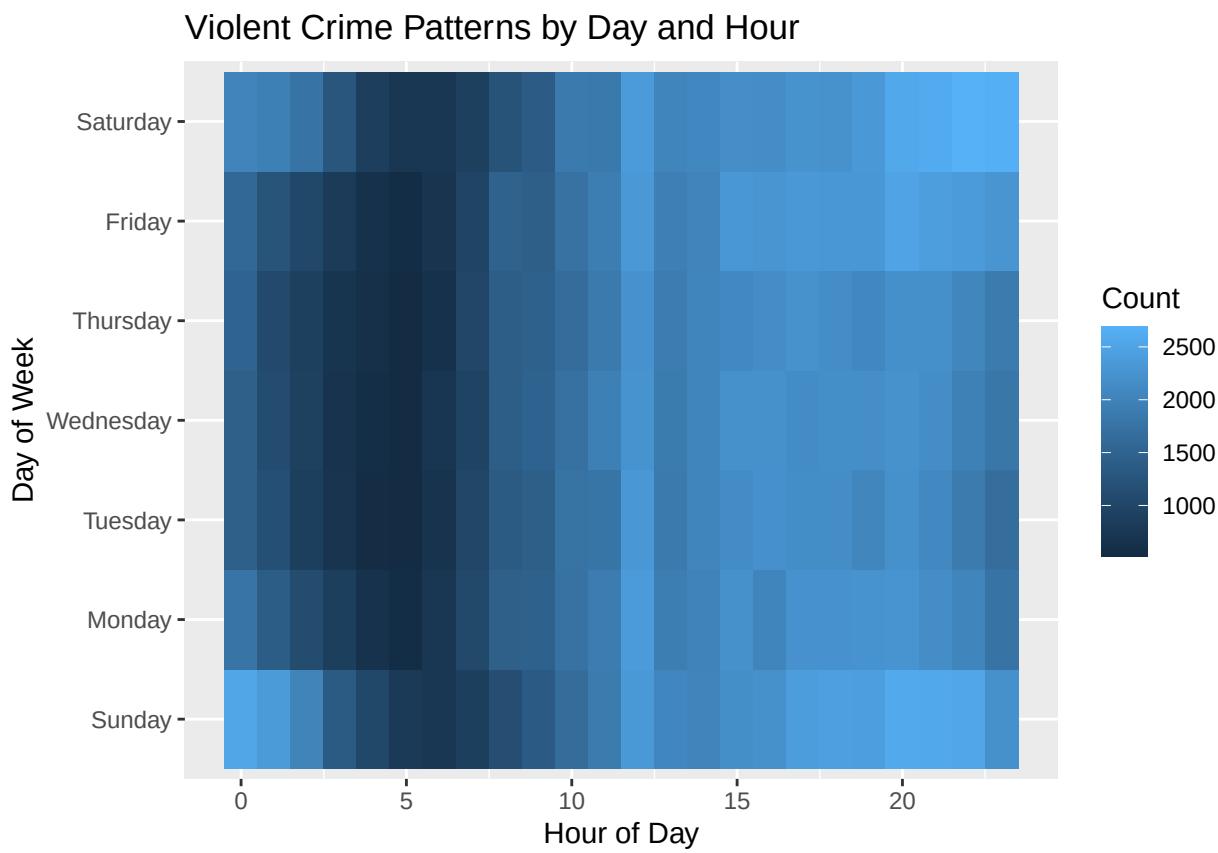
```r
  as.character(day_max$weekday),
  "with",
  day_max$n,
  "violent crimes\n"
)
```

```
## Highest: Sunday with 45599 violent crimes
```

```r
cat(
  "Lowest:",
  as.character(day_min$weekday),
  "with",
  day_min$n,
  "violent crimes\n"
)
```

```
## Lowest: Tuesday with 37981 violent crimes
```

```r
heat_data <- violent_df %>%
  count(weekday, hour)

ggplot(
  heat_data,
  aes(
    x = hour,
    y = weekday,
    fill = n
  )
) +
  geom_tile() +
  labs(
    title = "Violent Crime Patterns by Day and Hour",
    x = "Hour of Day",
    y = "Day of Week",
    fill = "Count"
  )
```




```
peak_times <- violent_df %>%
  count(weekday, hour) %>%
  group_by(weekday) %>%
  slice_max(n, n = 1) %>%
  ungroup()

peak_times
```

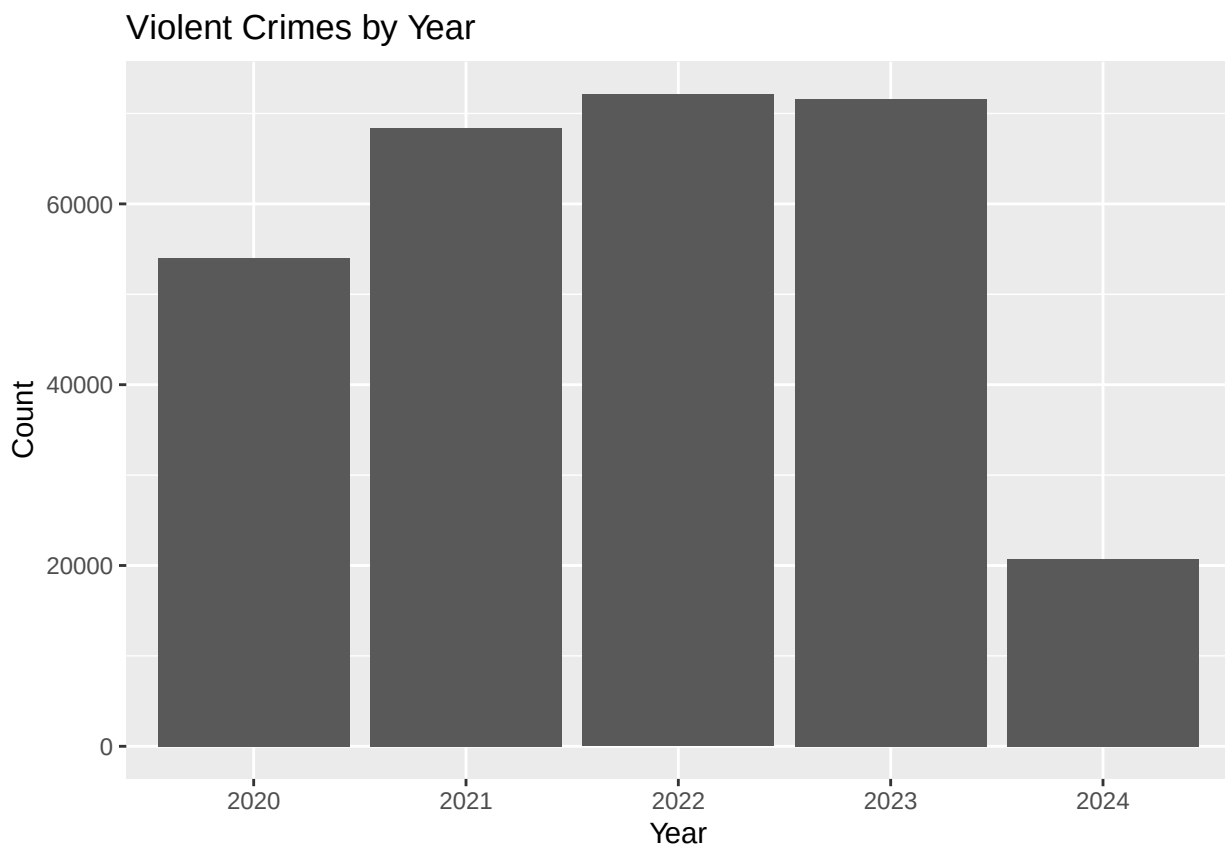
```
## # A tibble: 7 x 3
##   weekday    hour     n
##   <ord>     <dbl> <int>
## 1 Sunday       20  2585
## 2 Monday       12  2368
## 3 Tuesday      12  2301
## 4 Wednesday    12  2251
## 5 Thursday     12  2232
## 6 Friday       20  2502
## 7 Saturday     22  2692
```

```
df_2020 <- df[df$`DATE OCC` < as.POSIXct("2021-01-01"),]

df_2021 <- df[df$`DATE OCC` < as.POSIXct("2022-01-01") & df$`DATE OCC` >= as.POSIXct("2021-01-01"),]

df_2022 <- df[df$`DATE OCC` < as.POSIXct("2023-01-01") & df$`DATE OCC` >= as.POSIXct("2022-01-01"),]
```

```r
df_2023 <- df[df$`DATE OCC` < as.POSIXct("2024-01-01") & df$`DATE OCC` >= as.POSIXct("2023-01-01"),]

df_2024 <- df[df$`DATE OCC` < as.POSIXct("2025-01-01") & df$`DATE OCC` >= as.POSIXct("2024-01-01"),]
```

```r
violent_df2 <- df %>%
  mutate(
    crm_cd = as.numeric(`Crm Cd`)
  ) %>%
  filter(crm_cd %in% violent_codes) %>%
  mutate(

    date_only = as.Date(`DATE OCC`),

    year = year(date_only),

    month = month(
      date_only,
      label = TRUE,
      abbr = FALSE
    ),

    weekday = wday(
      date_only,
      label = TRUE,
      abbr = FALSE
    )
  )
```

```r
year_counts <- violent_df2 %>%
  count(year)

ggplot(year_counts,
       aes(x = factor(year), y = n)) +
  geom_col() +
  labs(
    title = "Violent Crimes by Year",
    x = "Year",
    y = "Count"
  )
```




```r
cat("2024 appears to have a lower crime rate due to missing data")
```

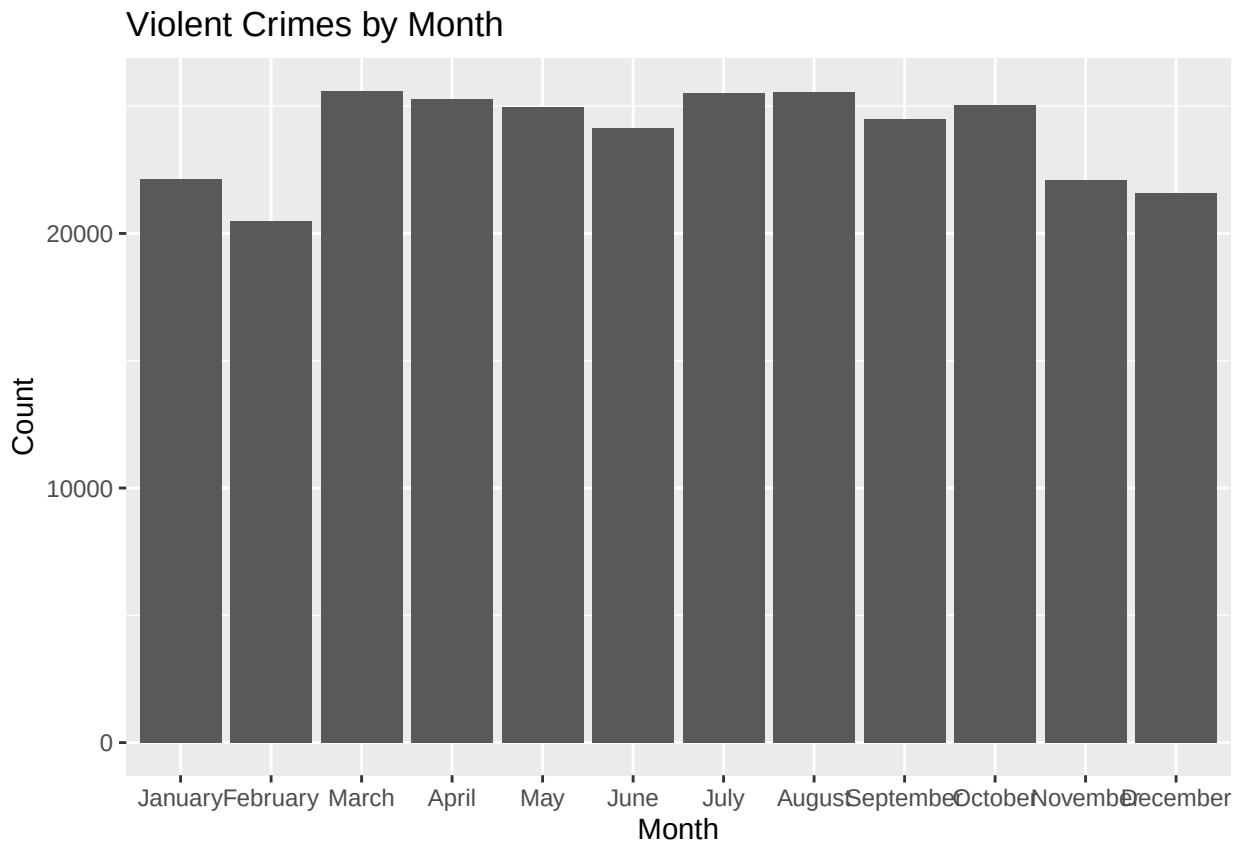
```
## 2024 appears to have a lower crime rate due to missing data
```

```r
year_counts %>%
  arrange(desc(n))
```

```
## # A tibble: 5 x 2
##    year     n
##   <dbl> <int>
## 1  2022 72119
## 2  2023 71604
## 3  2021 68406
## 4  2020 54030
## 5  2024 20739
```

```r
month_counts <- violent_df2 %>%
  count(month)

ggplot(month_counts,
       aes(x = month, y = n)) +
  geom_col() +
  labs(
    title = "Violent Crimes by Month",
```

```
    x = "Month",
    y = "Count"
  )
```




```
peak_days_by_year <- violent_df2 %>%
  count(year, weekday) %>%
  group_by(year) %>%
  slice_max(n, n = 1) %>%
  ungroup()

peak_days_by_year
```

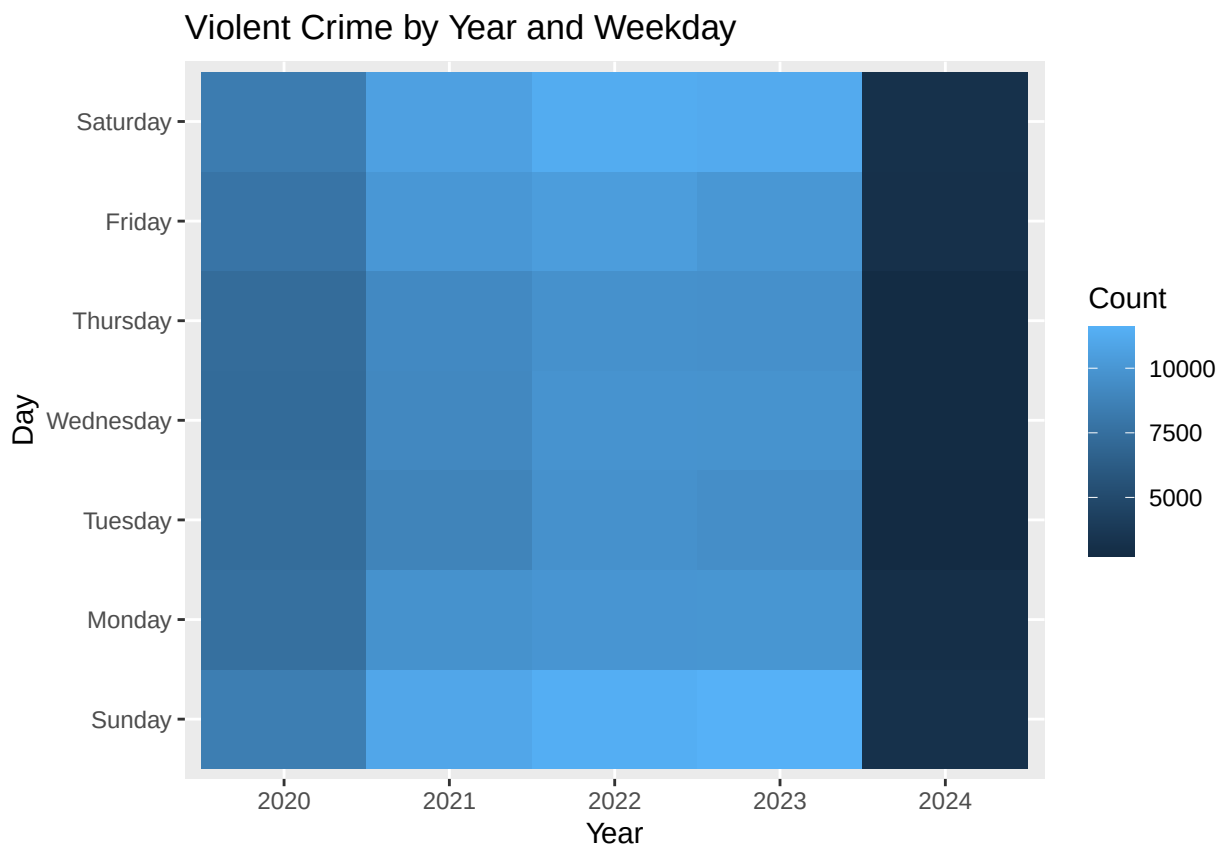
```
## # A tibble: 5 x 3
##    year weekday      n
##   <dbl> <ord>    <int>
## 1  2020 Sunday    8451
## 2  2021 Sunday   10962
## 3  2022 Sunday   11427
## 4  2023 Sunday   11612
## 5  2024 Saturday  3157
```

```
lowest_days_by_year <- violent_df2 %>%
  count(year, weekday) %>%
  group_by(year) %>%
```

```r
  slice_min(n, n = 1) %>%
  ungroup()

lowest_days_by_year
```

```
## # A tibble: 5 x 3
##    year weekday       n
##   <dbl> <ord>     <int>
## 1  2020 Wednesday  7233
## 2  2021 Tuesday    8817
## 3  2022 Thursday   9632
## 4  2023 Tuesday    9457
## 5  2024 Tuesday    2725
```

```r
heat_year_day <- violent_df2 %>%
  count(year, weekday)

ggplot(
  heat_year_day,
  aes(
    x = factor(year),
    y = weekday,
    fill = n
  )
) +
  geom_tile() +
  labs(
    title = "Violent Crime by Year and Weekday",
    x = "Year",
    y = "Day",
    fill = "Count"
  )
```




```r
weekday_year_table <- table(
  violent_df2$weekday,
  violent_df2$year
)

chisq.test(weekday_year_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  weekday_year_table
## X-squared = 60.968, df = 24, p-value = 4.658e-05
```

A Pearson's chi-squared test was conducted to examine whether the distribution of violent crimes across days of the week differed by year in Los Angeles. Results indicated a statistically significant association between year and day of week, χ 2 (24)=60.97,p<0.001, suggesting that while overall weekday crime patterns remained relatively consistent, the magnitude of violent crime across specific weekdays varied significantly across years.

```r
chi_result <- chisq.test(weekday_year_table)

chi2 <- as.numeric(chi_result$statistic)

n <- sum(weekday_year_table)

r <- nrow(weekday_year_table)
```

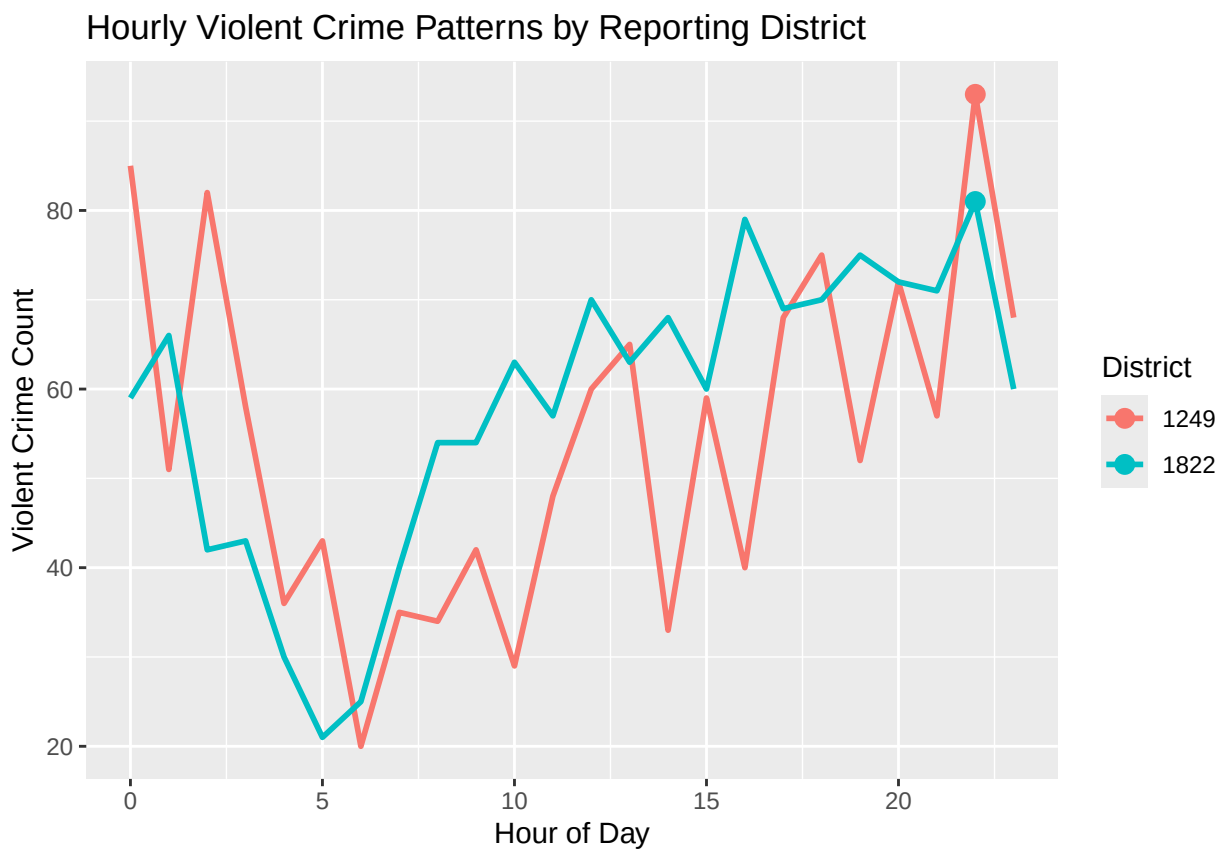
```r
c <- ncol(weekday_year_table)

cramers_v <- sqrt(
  chi2 / (n * min(r - 1, c - 1))
)

cramers_v
```

```
## [1] 0.007288811
```

Although weekday crime distributions differed significantly across years (p<.001), the effect size was small, suggesting that temporal violent crime patterns in Los Angeles remained relatively stable over time.

```r
top_districts <- c(645, 1249, 1822, 119)

time_location_df <- violent_df %>%
 filter(`Rpt Dist No` %in% top_districts) %>%
 count(`Rpt Dist No`, hour)
```

```r
peak_points <- time_location_df %>%
  group_by(`Rpt Dist No`) %>%
  slice_max(n, n = 1)

ggplot(
  time_location_df,
  aes(
    x = hour,
    y = n,
    color = factor(`Rpt Dist No`)
  )
) +
  geom_line(size = 1) +
  geom_point(
    data = peak_points,
    size = 3
  ) +
  labs(
    title = "Hourly Violent Crime Patterns by Reporting District",
    x = "Hour of Day",
    y = "Violent Crime Count",
    color = "District"
  )
```

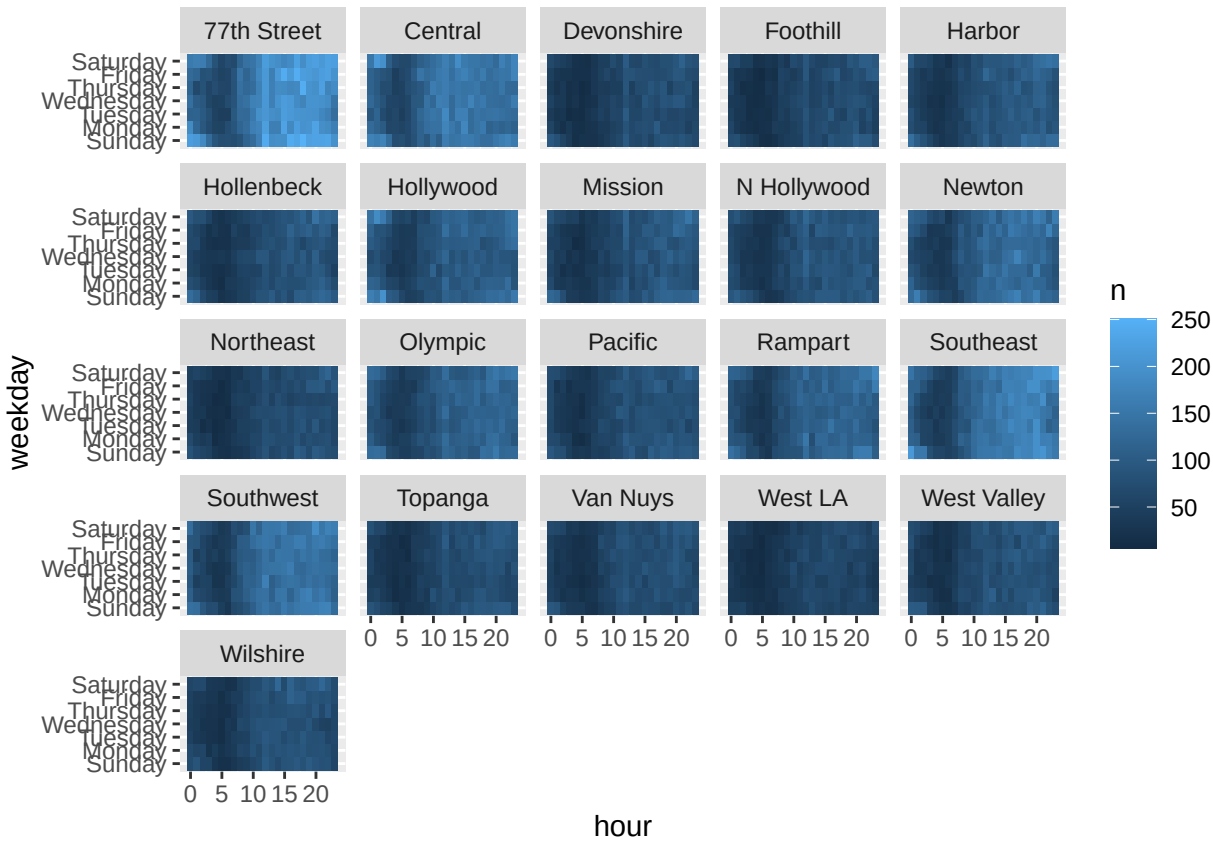



```
district_extremes <- time_location_df %>%
  group_by(`Rpt Dist No`) %>%
  summarise(
    peak_hour = hour[which.max(n)],
    peak_count = max(n),

    low_hour = hour[which.min(n)],
    low_count = min(n)
  )

district_extremes
```

```
## # A tibble: 2 x 5
##   `Rpt Dist No` peak_hour peak_count low_hour low_count
##   <chr>             <dbl>      <int>    <dbl>     <int>
## 1 1249                 22         93        6        20
## 2 1822                 22         81        5        21
```

```
heat_location <- violent_df %>%
 count(`AREA NAME`, weekday, hour)
ggplot(
 heat_location,
 aes(
   x = hour,
   y = weekday,
```

```
    fill = n
  )
) +
  geom_tile() +
  facet_wrap(~ `AREA NAME`)
```




```
area_extremes <- heat_location %>%
  group_by(`AREA NAME`) %>%
  summarise(
    peak_day = weekday[which.max(n)],
    peak_hour = hour[which.max(n)],
    peak_count = max(n),

    low_day = weekday[which.min(n)],
    low_hour = hour[which.min(n)],
    low_count = min(n)
  )

area_extremes
```

```
## # A tibble: 21 x 7
##    `AREA NAME` peak_day  peak_hour peak_count low_day   low_hour low_count
##    <chr>       <ord>         <dbl>      <int> <ord>        <dbl>     <int>
##  1 77th Street Sunday           18        251 Tuesday          5        46
##  2 Central     Saturday          1        210 Tuesday          5        45
```

```
##   3 Devonshire  Saturday         21         111 Tuesday          3         10
##   4 Foothill    Sunday           20         115 Friday           5          6
##   5 Harbor      Saturday         22         145 Wednesday        5         16
##   6 Hollenbeck  Saturday         20         142 Monday           5         14
##   7 Hollywood   Sunday            2         196 Tuesday          5         28
##   8 Mission     Saturday         22         149 Thursday         4         11
##   9 N Hollywood Saturday         12         122 Monday           5         19
##  10 Newton      Wednesday        17         173 Thursday         5         28
## # i 11 more rows
```

```
top_hotspots <- heat_location %>%
  arrange(desc(n)) %>%
  slice(1:5)

top_hotspots
```

```
## # A tibble: 5 x 4
##   `AREA NAME` weekday   hour     n
##   <chr>       <ord>    <dbl> <int>
## 1 77th Street Sunday      18   251
## 2 77th Street Thursday    18   243
## 3 77th Street Friday      17   243
## 4 77th Street Friday      15   242
## 5 77th Street Sunday      16   241
```